



FFF Printing Fundamentals

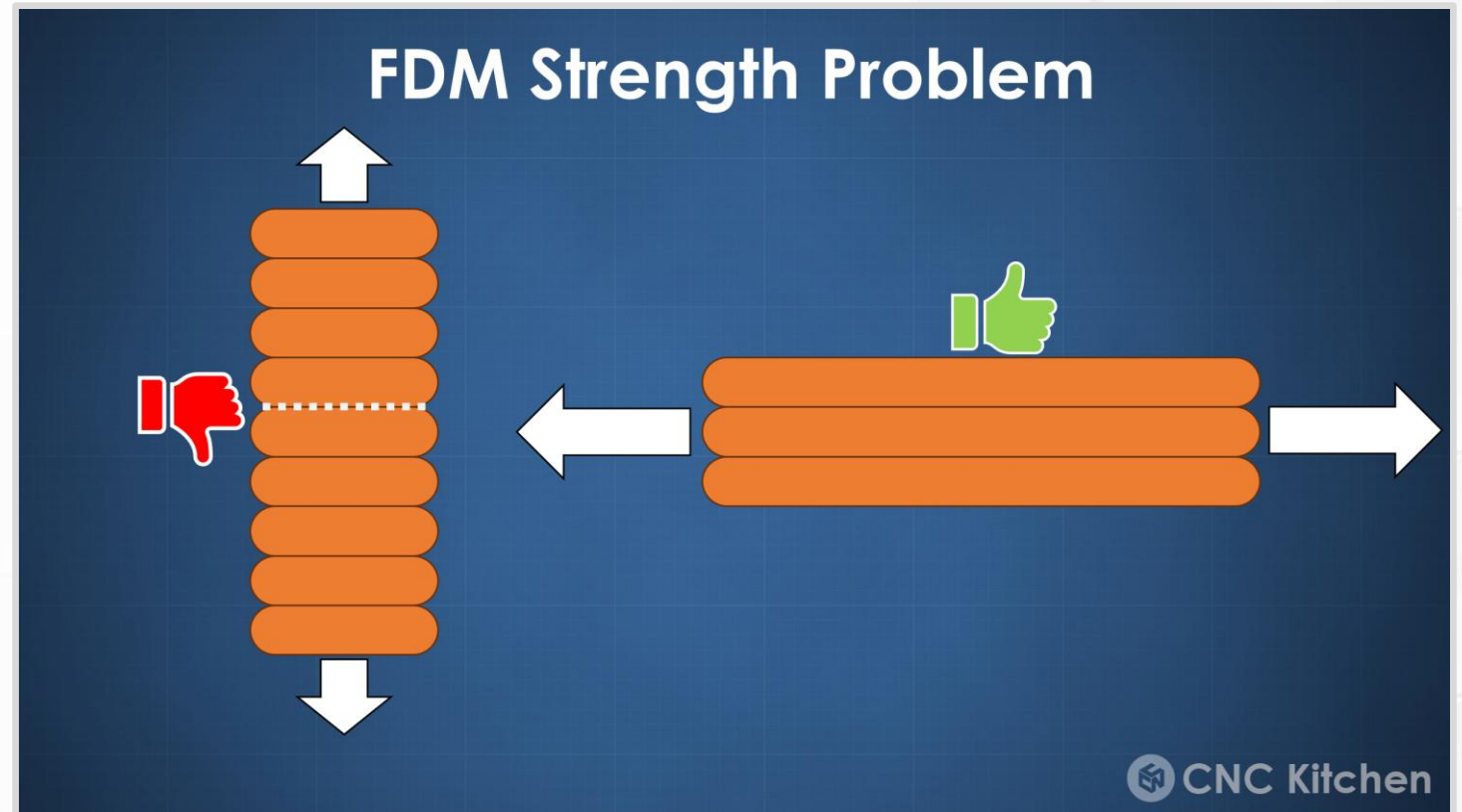
Part 2: Limitations

Thermal Cycle

- Some plastics shrink as they cool.
 - Up to 1 or 2%.
 - Seems small, but large enough to distort dimensions.
 - Leads to cracking and delamination in extreme cases.
- Advice?
 - Tightly control ambient temperature and cooling rate.
 - Some filaments (ABS, Nylon) like it hot → Extra heating.
 - Some filaments (PLA) like it cold → Extra cooling.
 - PETG is somewhere in the middle.

Anisotropic Strength

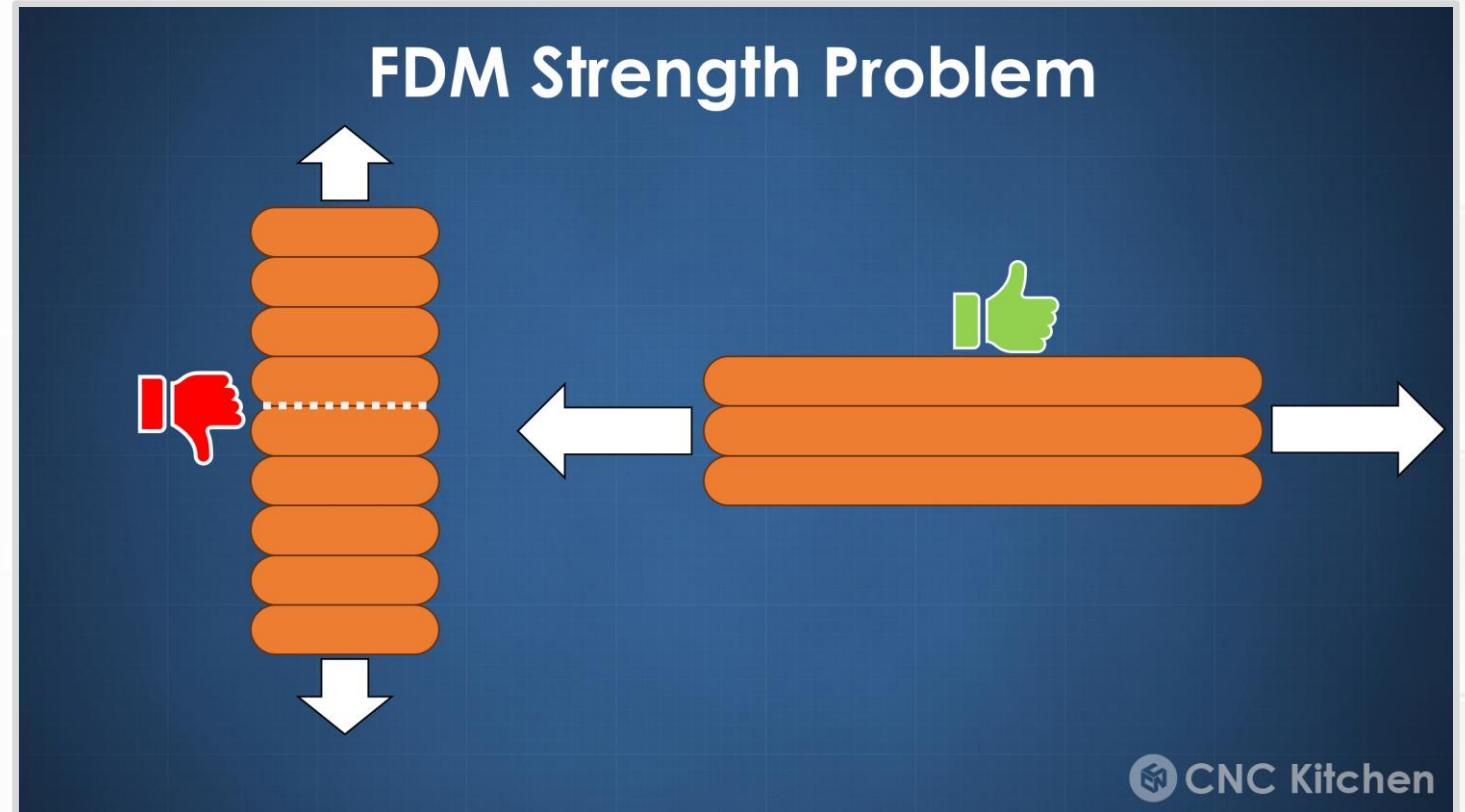
- Print orientation and layer lines affect strength.
- Layer-to-layer bonding is a weak point!
- Within an extrusion, hot filament bonds to hot filament.
- For a new layer, hot filament bonds to a warm layer.



(Picture from [CNC Kitchen](#))

Anisotropic Strength

- The “wrong” orientation can lose 50% of strength.
- Advice?
- Avoid tall and skinny shapes.
- Improve layer bonding.
 - Increase printing temps.
 - Decrease printing speed.



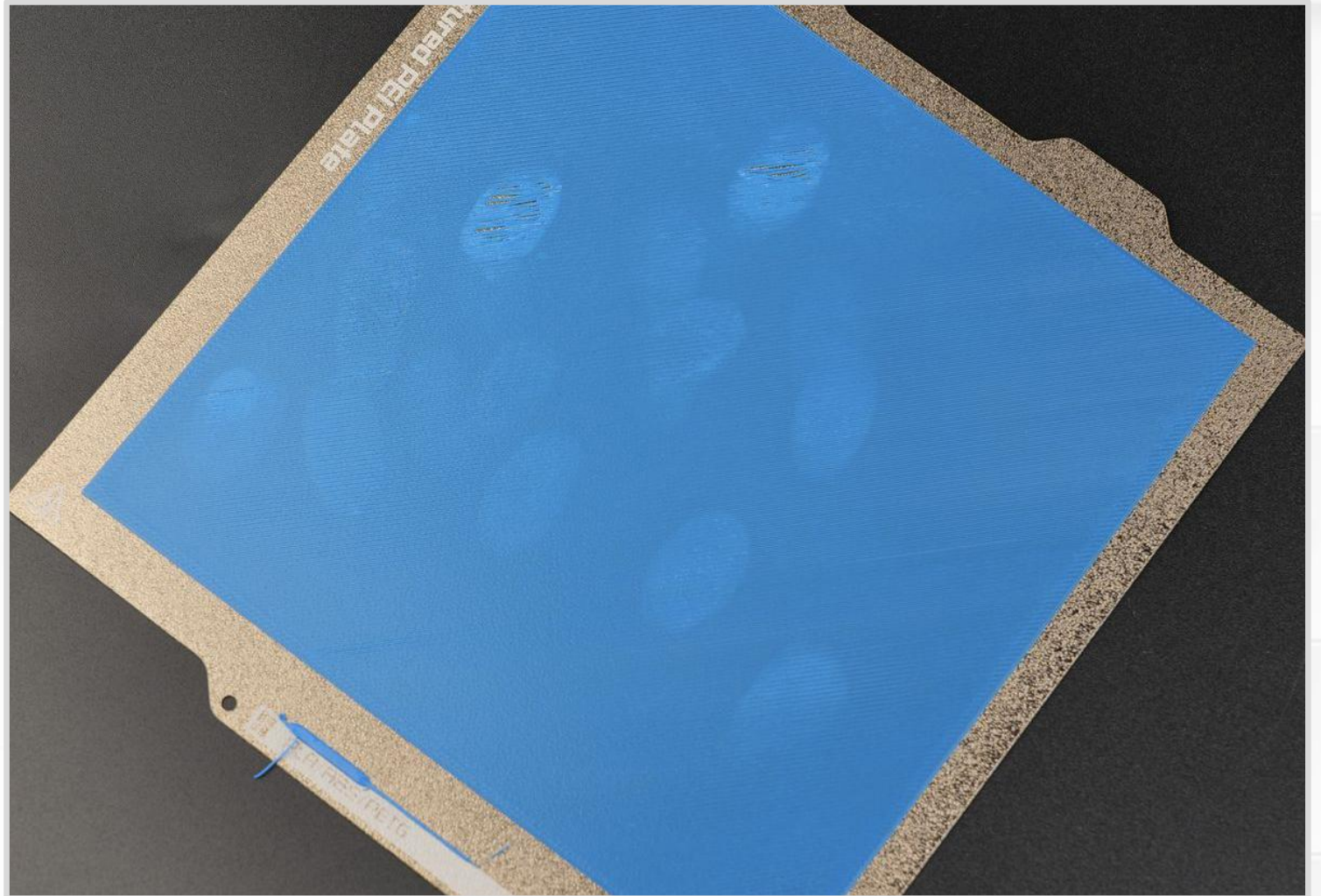
(Picture from [CNC Kitchen](#))

First Layer

- What is the most important layer in a 3D print?
 - The first layer!
 - First layers keep everything locked in place.
- Advice?
 - **Always watch the first layer.**
 - Tune first layer squish.
 - Flat surface area on the bottom of the 3D object.
 - Heated print bed.
 - Perfectly clean print bed.
 - Apply adhesion promoters (glue).

First Layer Mistakes

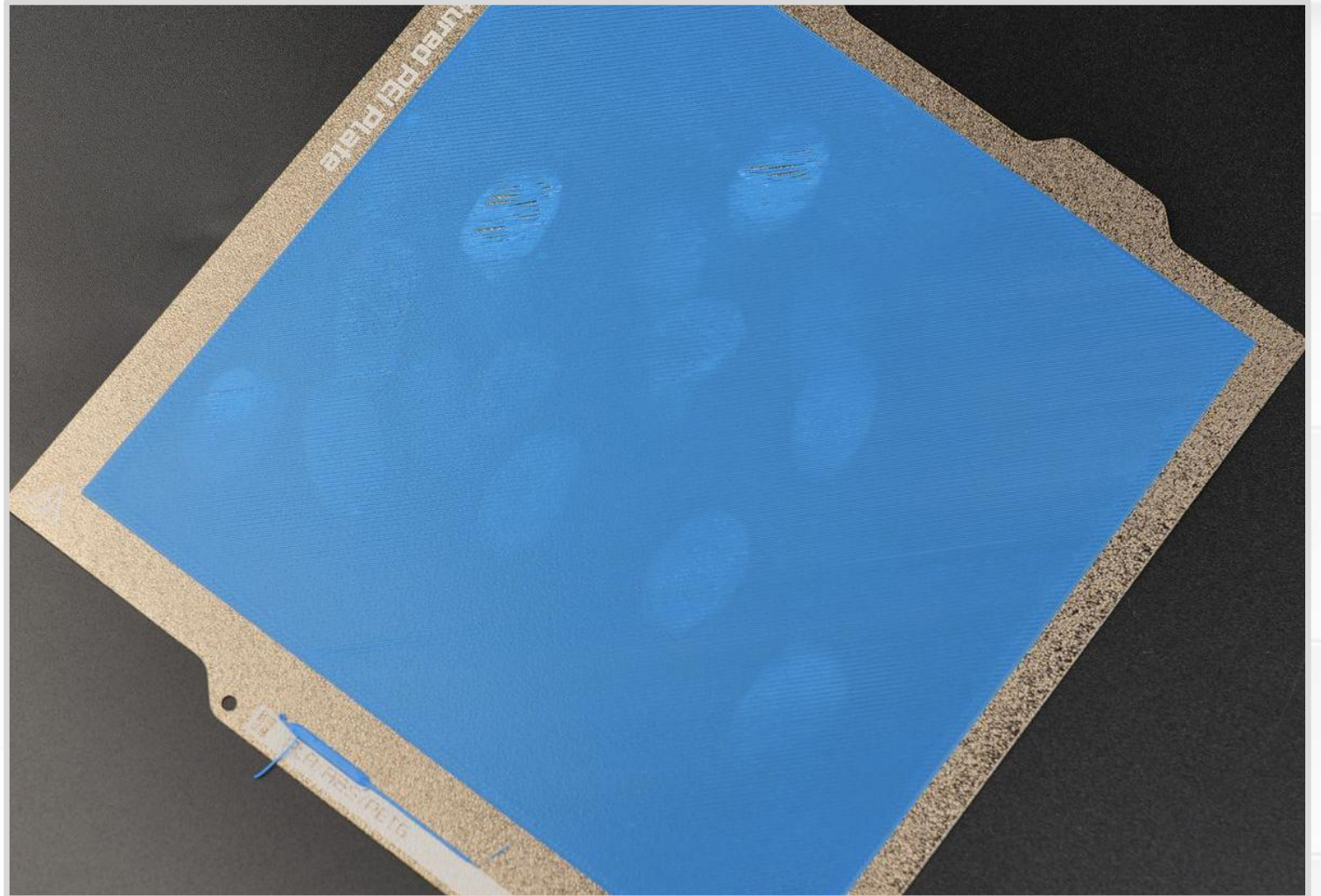
- What is wrong with this picture?
- **Fingerprints!** Plastic does not stick to oily fingerprints.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

- Advice?
- Deep cleaning: dish soap, soft brush, and water.
- Everyday use: window cleaner with ammonia.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

- What happens when a first layer does not stick?
- The print breaks free and makes “spaghetti”.
- You can see when things went wrong.
- Some of the rectangular print finished.



(Screenshot from [Everything STEM](#))

Stair-Stepping

- Printing with layers → Vertical slopes turn into stair steps.
- Mostly cosmetic effect.
- More pronounced on shallow angles.
- Advice?
- Rotate objects to hide layer lines.
- Smaller layer height at the cost of increased print time.



(Picture from [Hamburg University Research Project](#))

First Layer Warping

- Why do corners and edges peel up?
- Regions shrink at different rates → Internal stresses → Stresses concentrate in corners.
- Shrinkage is a percent → Large parts have the most trouble.



(Picture from [Creality Cloud Blog](#))

First Layer Warping

Consider this analogy.

- Imagine you want to remove a label or piece of tape.
- What do you do?
- Start peeling up the corner because that is the weak point.
- Same thing here.
- Physics found the weak point.



(Picture from [Crealitty Cloud Blog](#))

First Layer Warping

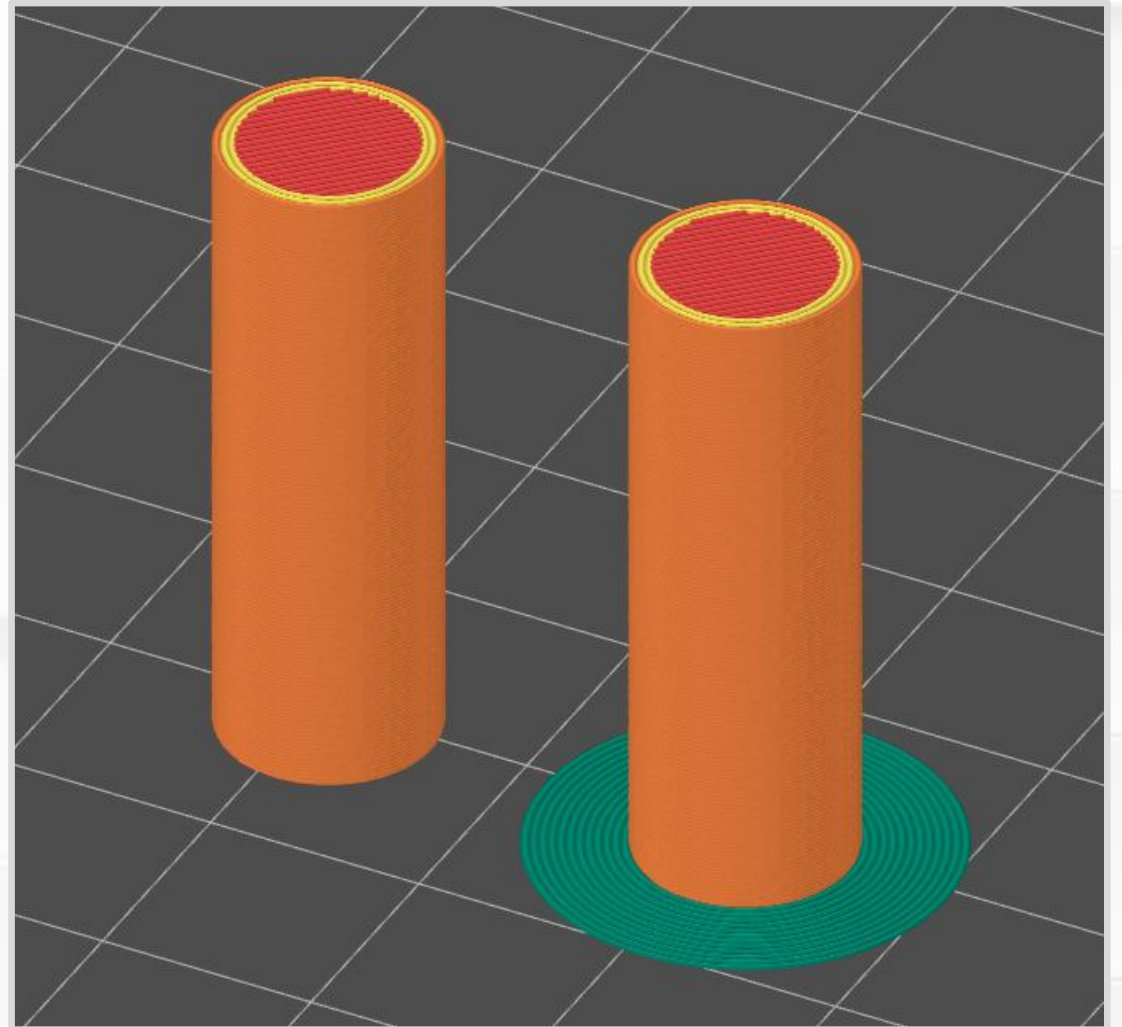
- Advice?
- Better control over temperatures.
 - Filament temperature.
 - Print chamber temperature.
 - Cooling fans.
- Evenly distributed cooling.
 - No fans or drafts that cool one side.
- Round corners to reduce stress concentration.



(Picture from [Crealitty Cloud Blog](#))

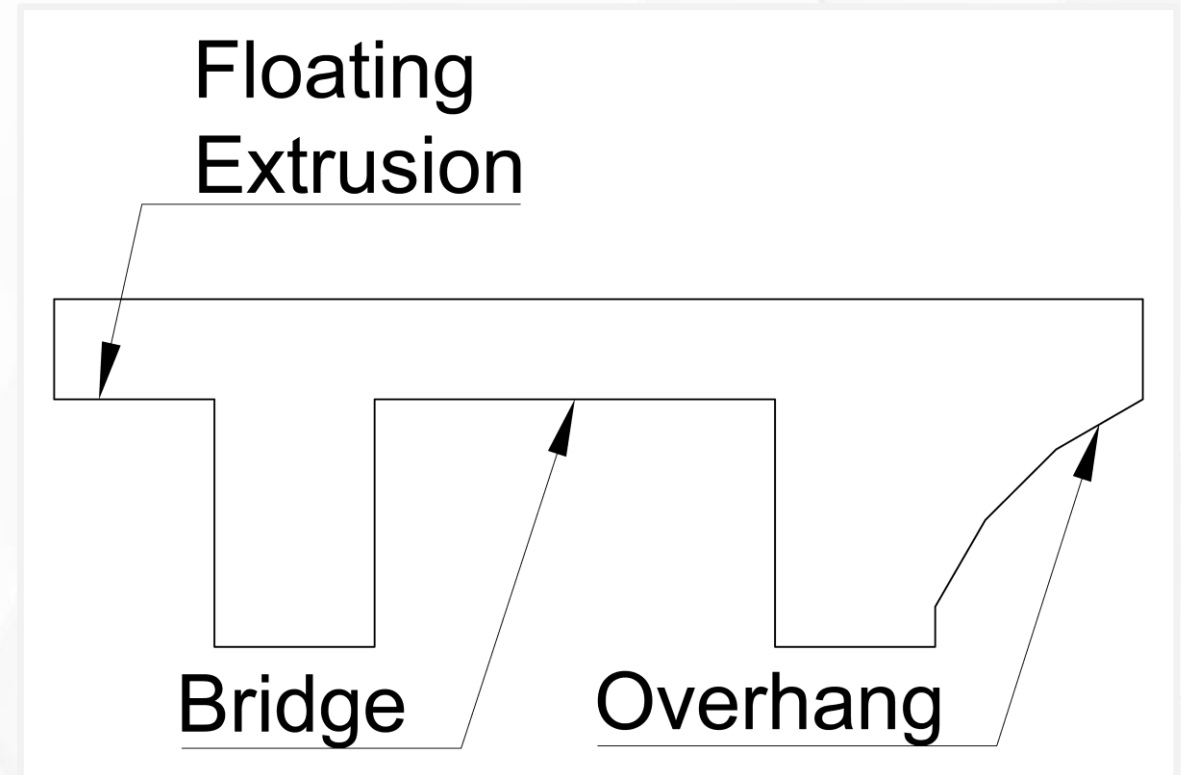
Brim

- Extra bottom surface area.
- Fights against first layer warping.
- Concentric extrusions joined to bottom perimeter. (Green lines on the right.)
- Added with a slicer.
- One layer tall.
- Easily peeled away after printing.



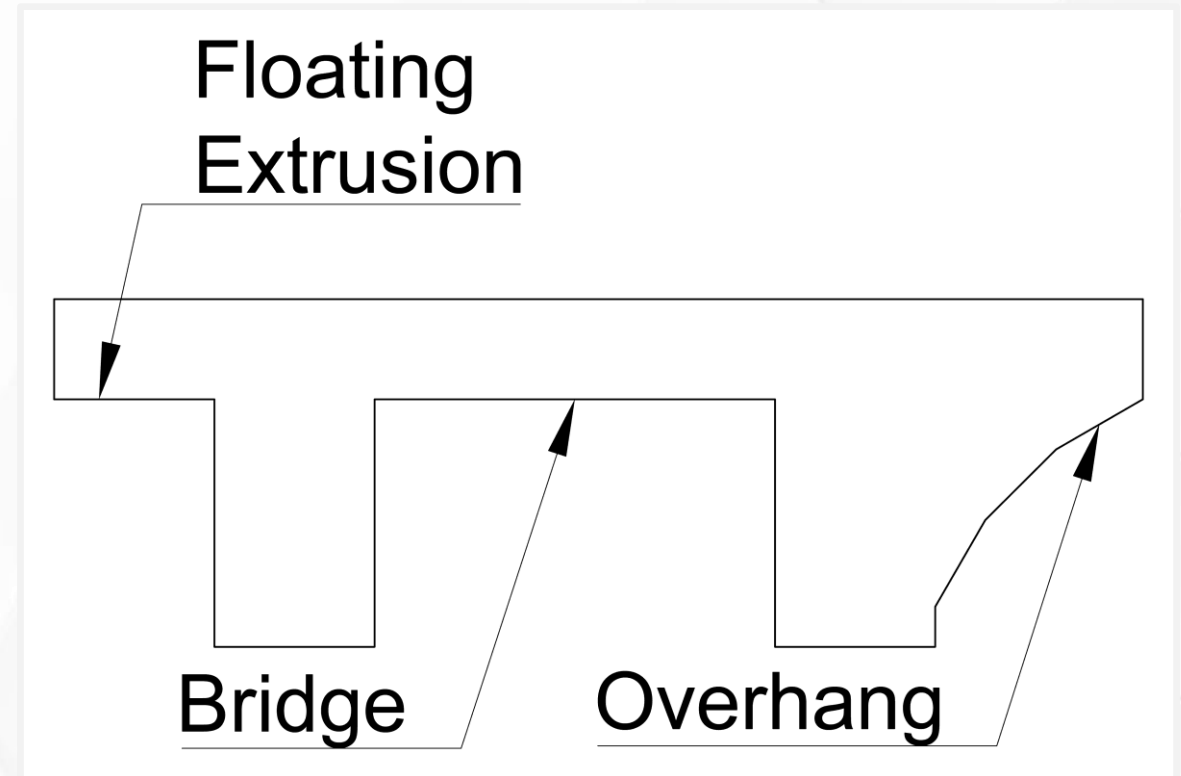
Levitating Extrusions

- *Critical thinking:* What happens when you try to levitate extrusions?
- Extrusions work best when you press them onto a solid surface.
- Predict what this print will look like.
- What parts will be the most deformed?



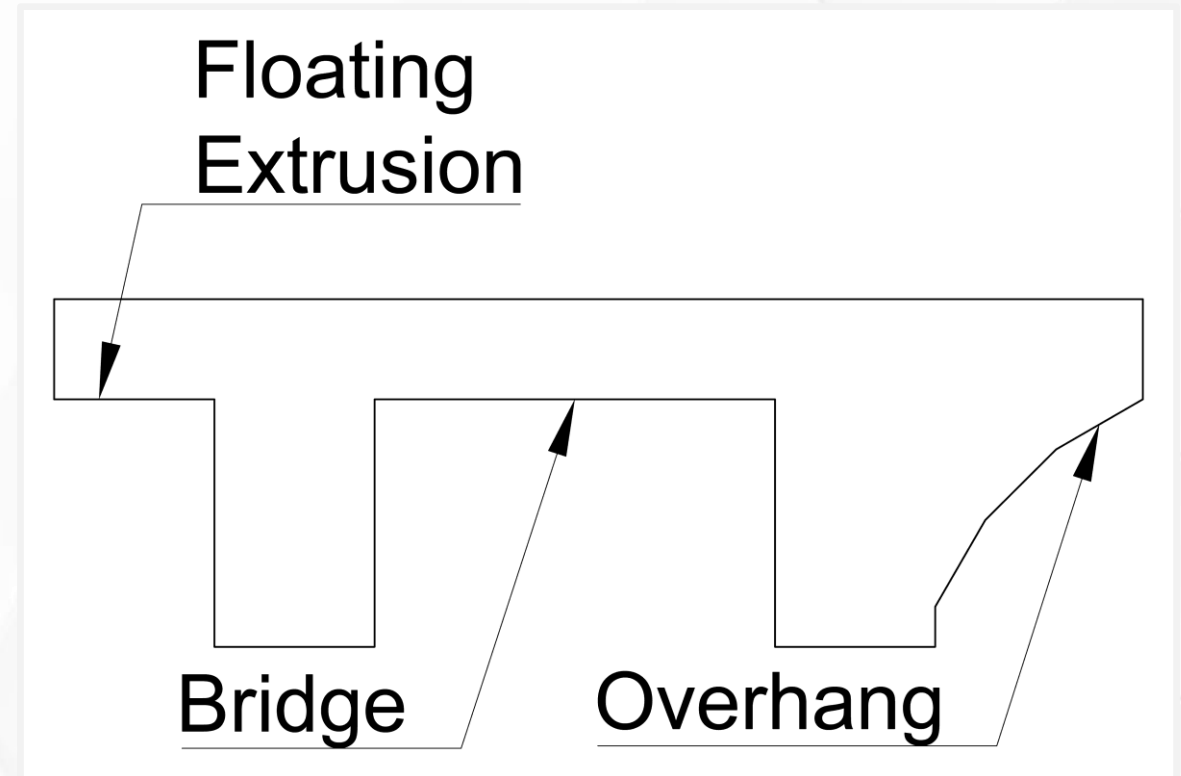
Overhang

- Extrusions that lean over an edge.
- Only the outermost extrusion is affected.
- Partially supported along one side.
- Advice?
- Keep overhangs at 45° to be safe.



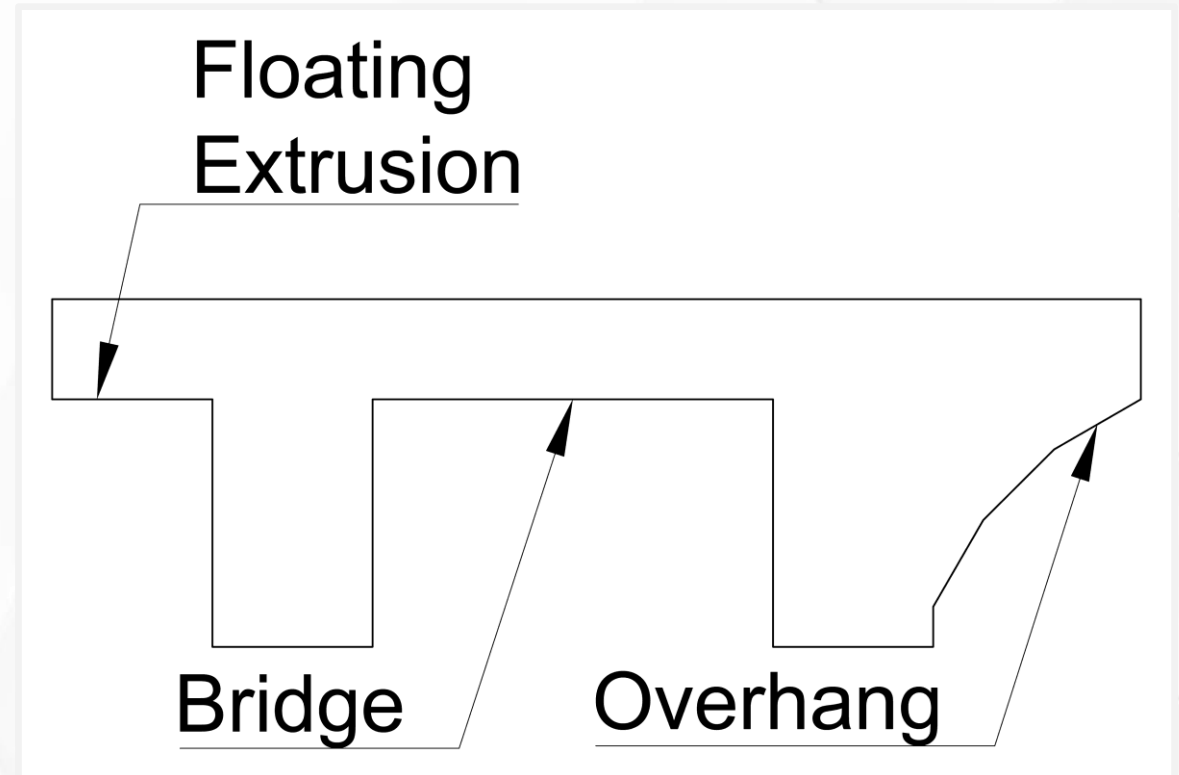
Bridge

- Both ends of an extrusion are supported and anchored.
- Middle hovers over a gap.
- The first few layers sag and droop down.
- Eventually recovers to flat layers.
- Advice?
- Keep bridges as short as possible.



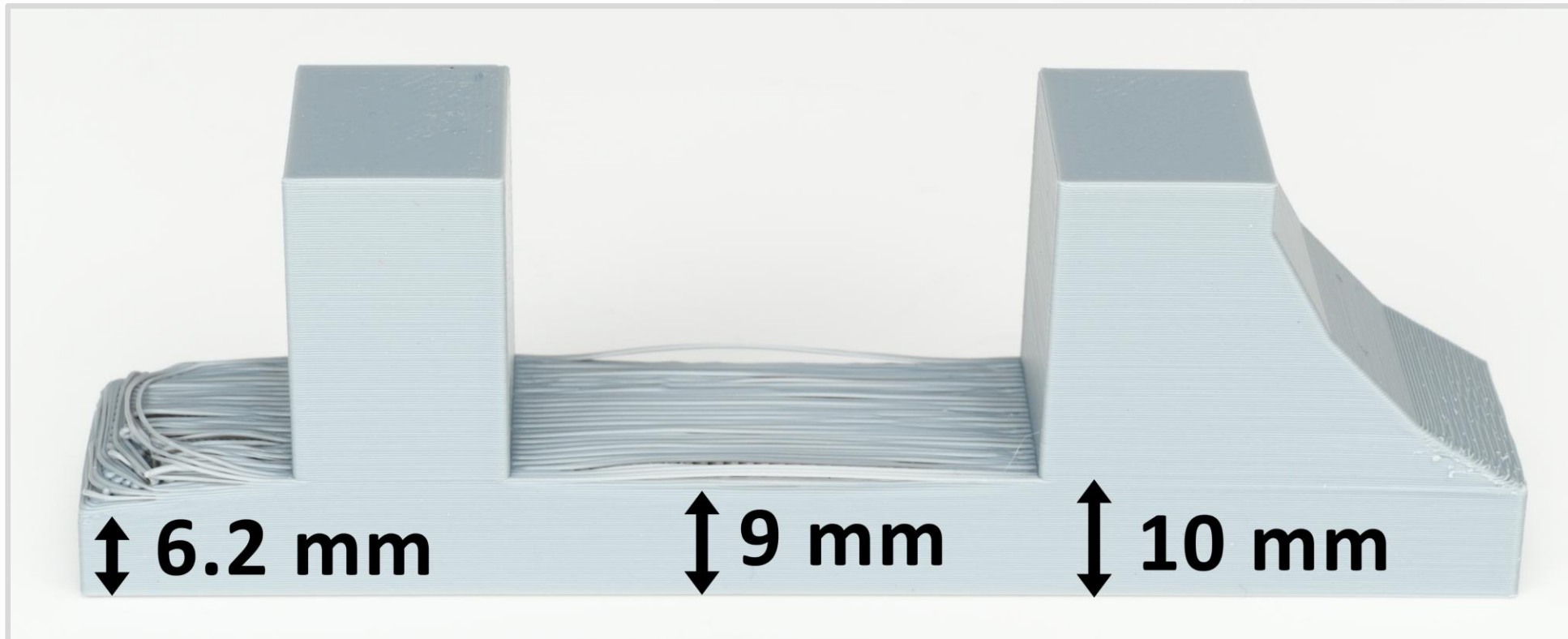
Floating Extrusion

- Extrusions that hang in the air.
- At least one end is not anchored to a solid surface.
- Advice?
- **Avoid. Very unreliable.**



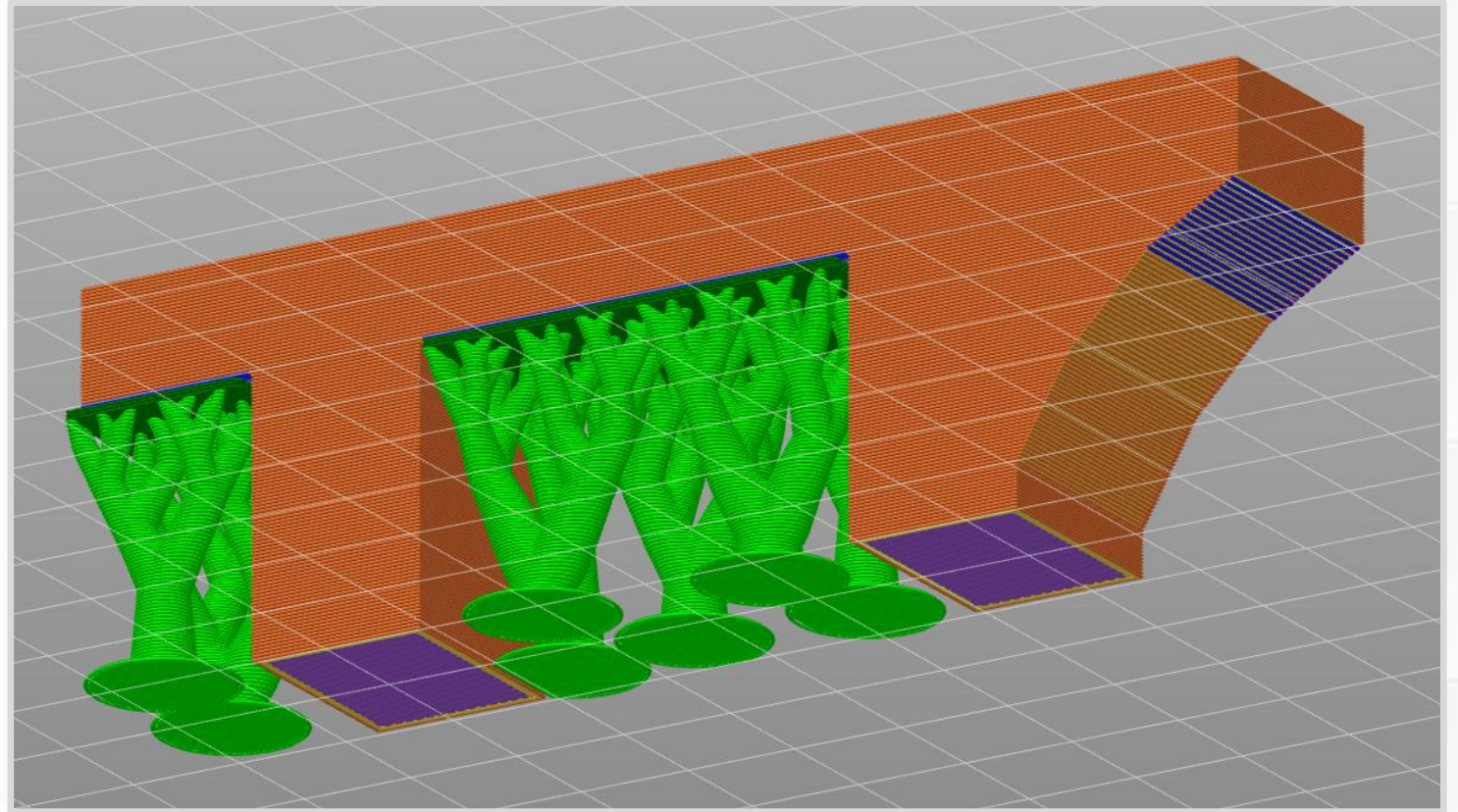
Stress Test Results

- All overhang angles look good.
- Bridge recovers to flat layers in 1 mm (**5 layers**).
- Floating extrusions recover in 3.8 mm (**19 layers**).



Supports

- Temporary scaffold that fixes:
 - Floating extrusions.
 - Long bridges.
 - Steep overhangs.
- Added by the slicer.
(Green trees-like branches.)
- Extra complexity, plastic, and print time.

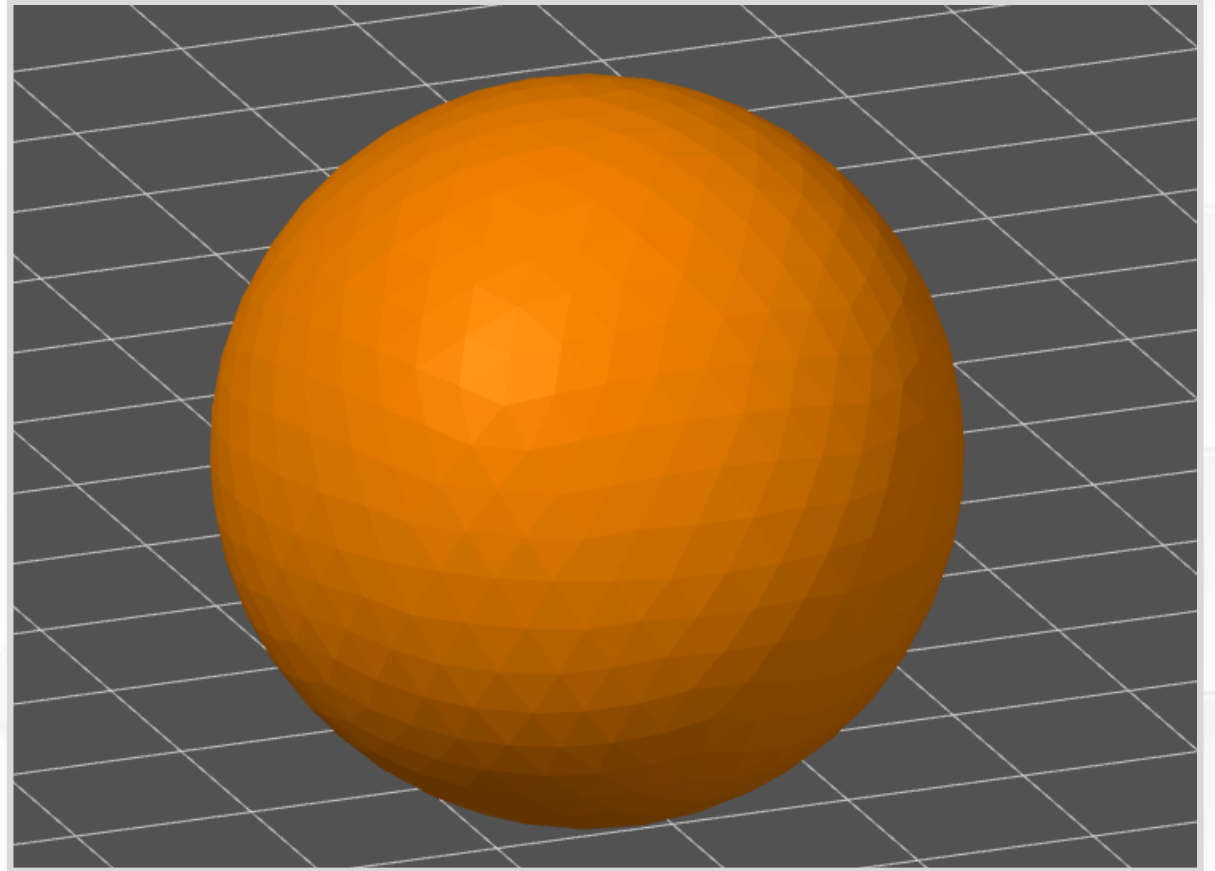


Supports

- In special situations, you can include supports as part of a 3D model.
- Tree/organic supports can reach into difficult spaces.
- Supports are a delicate balance. Why?
 - If they do not bond well enough ... they hold nothing in place.
 - If they bond too well ... they become part of the print.
- Supports can leave a rough surface finish.
- Sometimes you need to cut away pieces of support.

Limitations Review

- Printing a ball runs into some FFF limitations.
 - Very bad **first layer contact**. Huge spaghetti risk.
 - Steep overhangs on the bottom.
 - Second layer is steep enough to be a floating extrusion.
 - Stair-stepping creates uneven surface finish.



Limitations Review

How can we fix this print?

- Cannot fix stair-stepping by changing print orientation.
 - Use small layer heights to reduce visual artifacts.
 - Use adaptive layer heights to make the surface look more uniform.
- Brim is marginally helpful.
 - Object starts with too little surface area.
- Add supports for most of the bottom.
- Print two halves, then put them together.